

1. The number of students infected with flu at a high school after t

days is modeled by the function $P(t) = \frac{500}{1 + 99e^{-0.2t}}$.

The school will close when 400 of the 500 students are infected.

After how many days will the school close?

days

(Round to the nearest whole number.)

2. The logistic growth function $P(x) = \frac{90}{1 + 271e^{-0.122x}}$ models the percentage, $P(x)$, of people in a certain country who are x years old and have some coronary heart disease. Use this function to find the age at which the percentage of people who have some coronary heart disease is 58%.

What is the age at which 58% of people have some coronary heart disease?

years old (Round to the nearest integer.)

3. The spread of a highly contagious virus in a high school can be described by the logistic function

$$y = \frac{6000}{1 + 999e^{-0.9x}}$$

where x is the number of days after the virus is identified in the school and y is the total number of people who are infected by the virus.

(a) What is the upper limit of the number infected by the virus during this period?

(b) How much time has gone by when 4500 people are infected?

(Round to three decimal places)

(c) How many people are infected when the disease is first diagnosed?

(d) If you round the number of infect people to the nearest whole number, how long will it take for the maximum number of people to be infected **(Round the number of days up to a whole number)**

(a) The upper limit of the number infected by the virus during the period is .

(b) The number of days that have passed by when 4500 people are infected is days

(Round to three decimal places)

(c) On the day of first diagnosis there are already infections

(d) Rounding to the nearest whole number, it will take days for the maximum number of infections to be reached **(Round the number of days up to a whole number)**